

**Predicting Poor Mental Health
Days Using Behavioral and
Demographic Data from the
Behavioral Risk Factor
Surveillance
System (BRFSS), 2022–2023**

SENIOR THESIS

INFO-I 492 – Senior
Thesis (Fall 2025)

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Introduction / Background

Mental Health Importance

Mental health issues affect millions of U.S. adults and create major social and economic burdens.

Why BRFSS

- BRFSS is the largest U.S. health survey (~400k+ adults/year).
- Contains behavioral + demographic + mental health indicators.

Your research goal

Predict “poor mental health days”
using lifestyle and
socioeconomic variables.

Research Question:

Which behavioral and socioeconomic factors best predict poor mental health days in U.S. adults using BRFSS 2022–2023 data?

How does education modify or interact with income in shaping mental health outcomes?

Dataset Description

- **Dataset: BRFSS 2022–2023 (Centers for Disease Control and Prevention)**
- **Sample size: ~400,000 U.S. adults**
- **Variables used:**
 - **Poor mental health days**
 - **Exercise**
 - **Smoking**
 - **Alcohol use**
 - **Education**
 - **Income**
 - **Age group**
 - **Sex**

Methodology

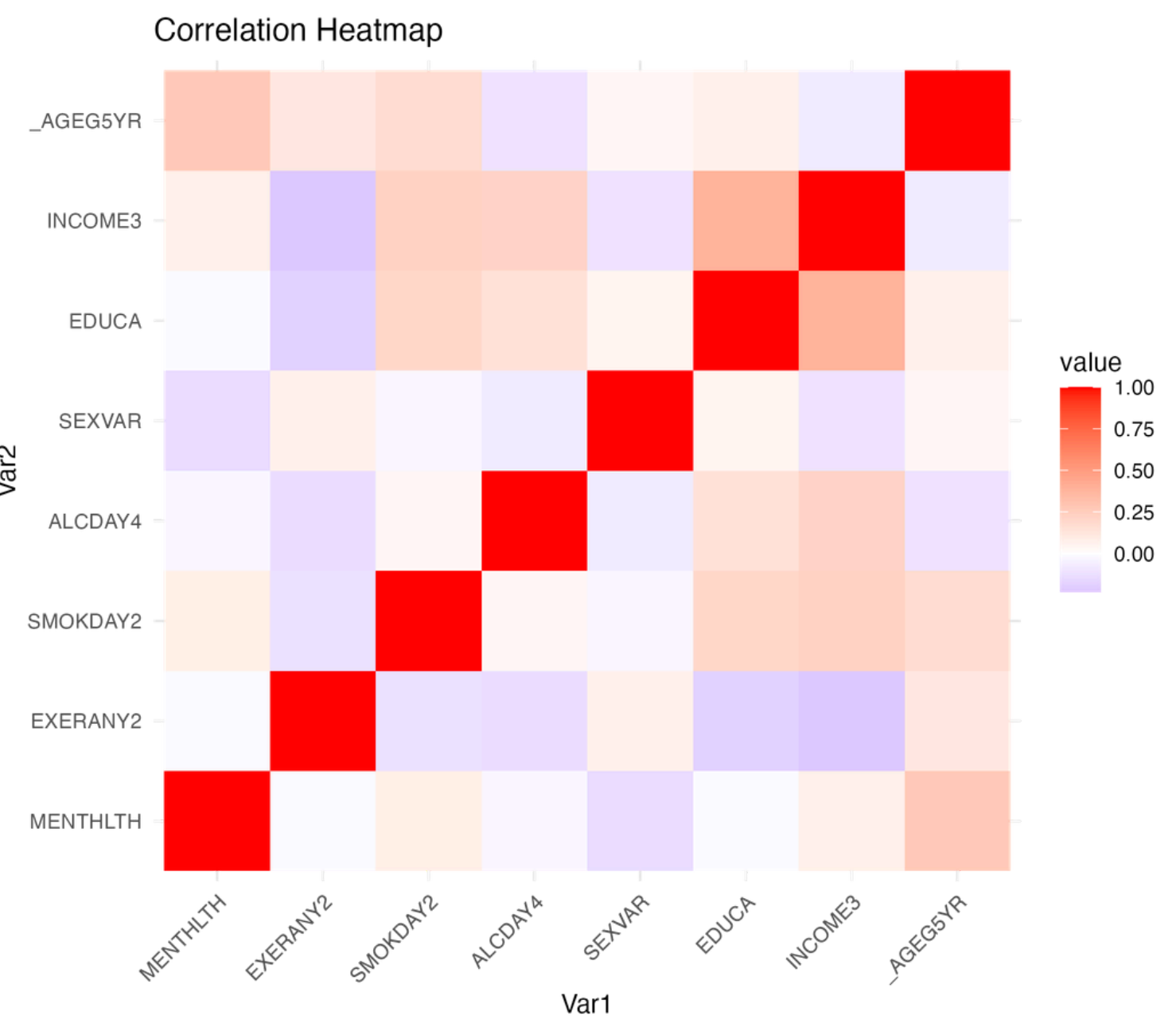
- Data cleaning
- Variable recoding
- Regression models
- Interaction term (education × income)
- ROC/AUC evaluation

Stratified analysis by sex and age groups

Key Visualization #1

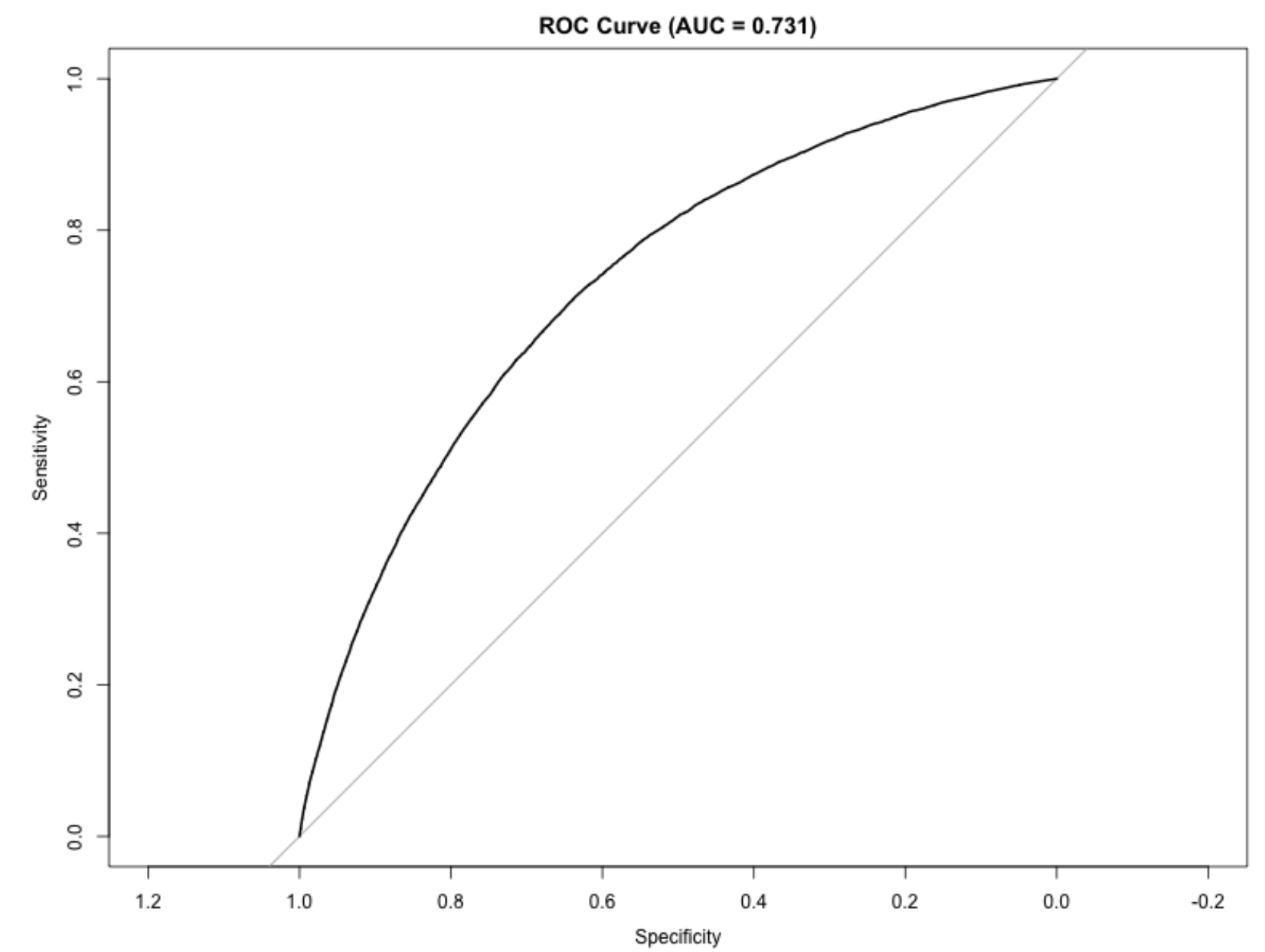
Correlation heatmap

Shows relationships among predictors
(education, income, exercise, smoking, alcohol)



ROC curve

Logistic model achieved
moderate predictive ability
(AUC = 0.731).

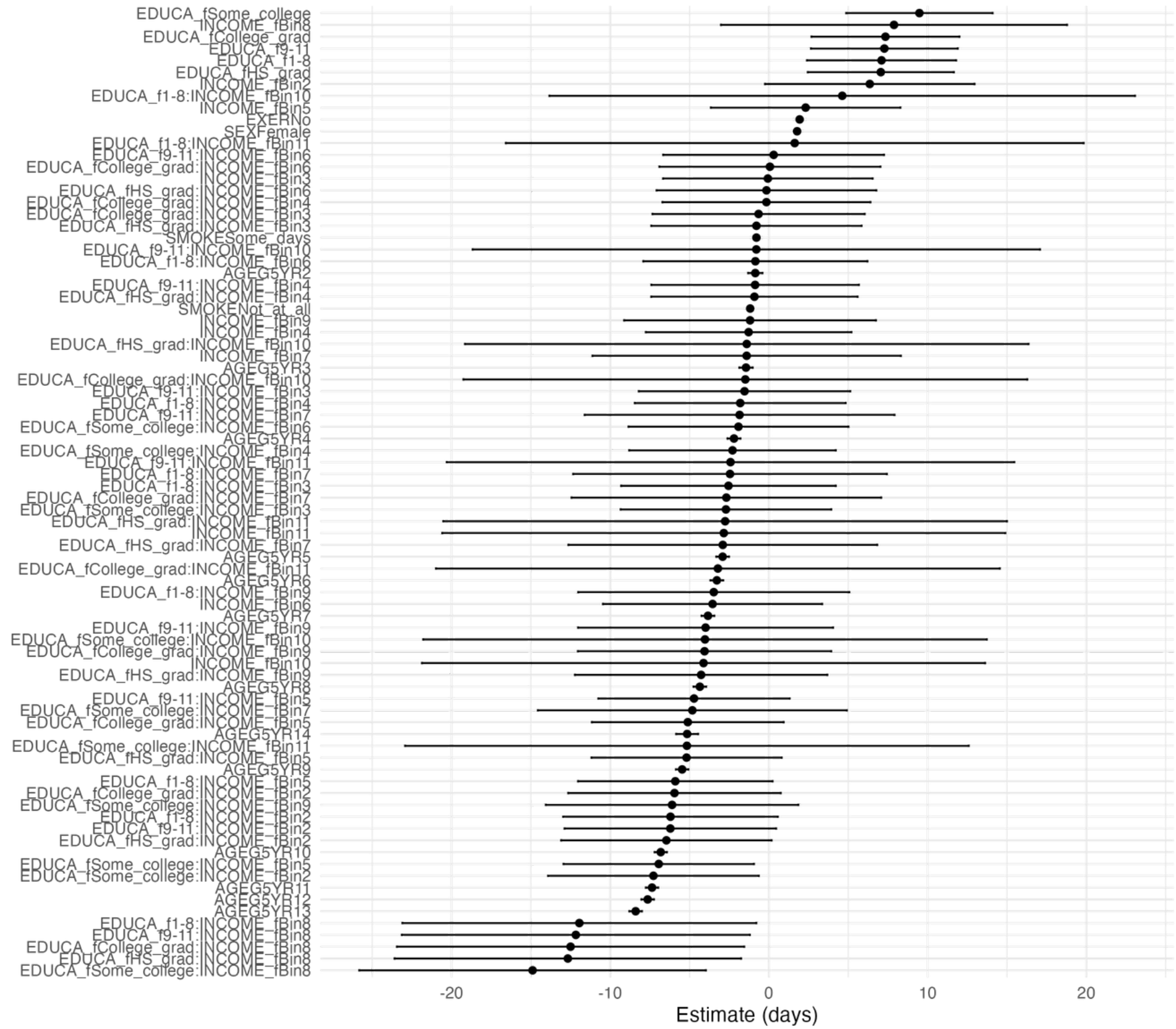


Key Visualization #2 Coefficient plot

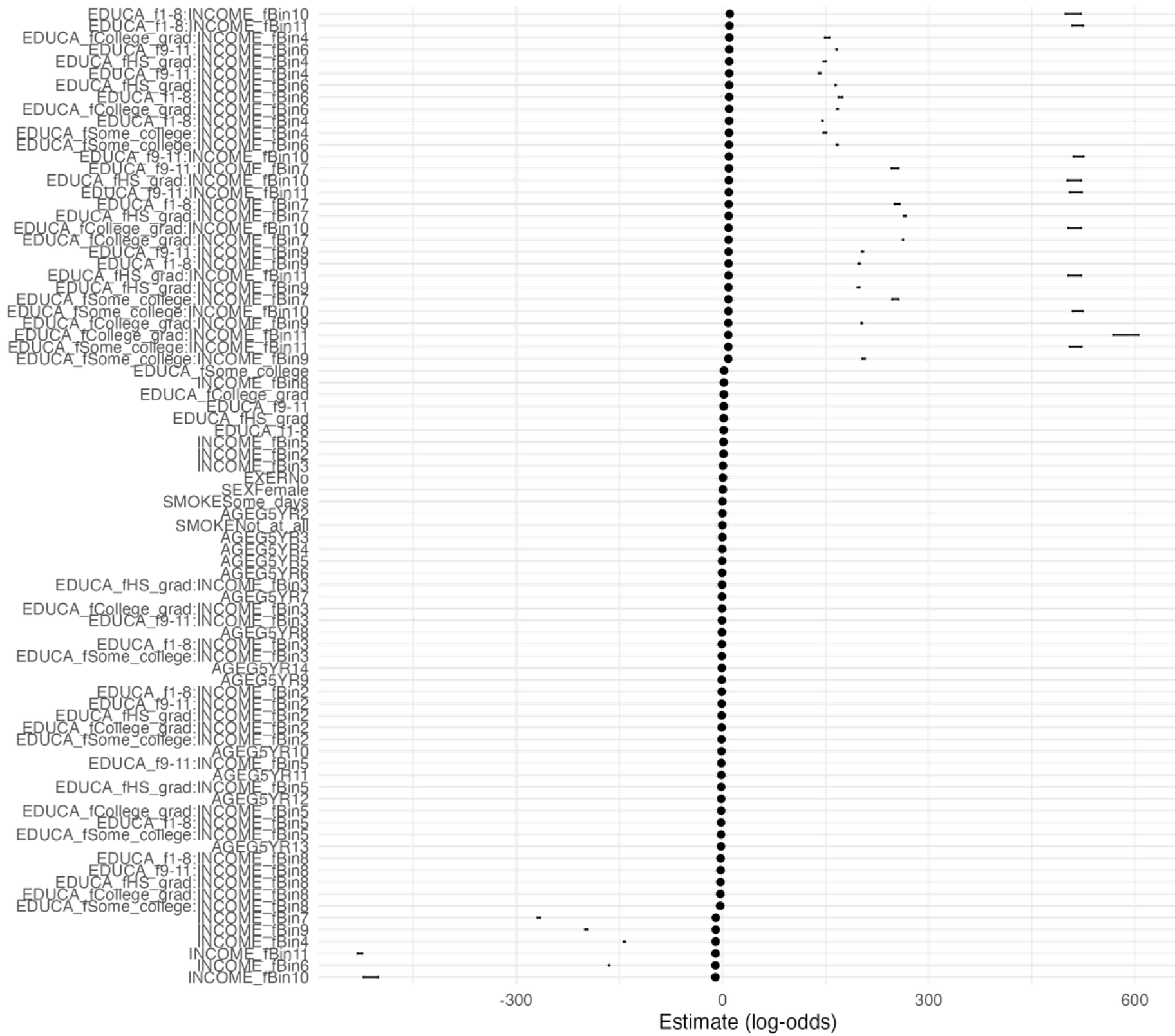
Higher education levels reduce predicted poor mental health days.

Smoking, alcohol use, and low income increase predicted mental distress.

Linear Model Coefficients



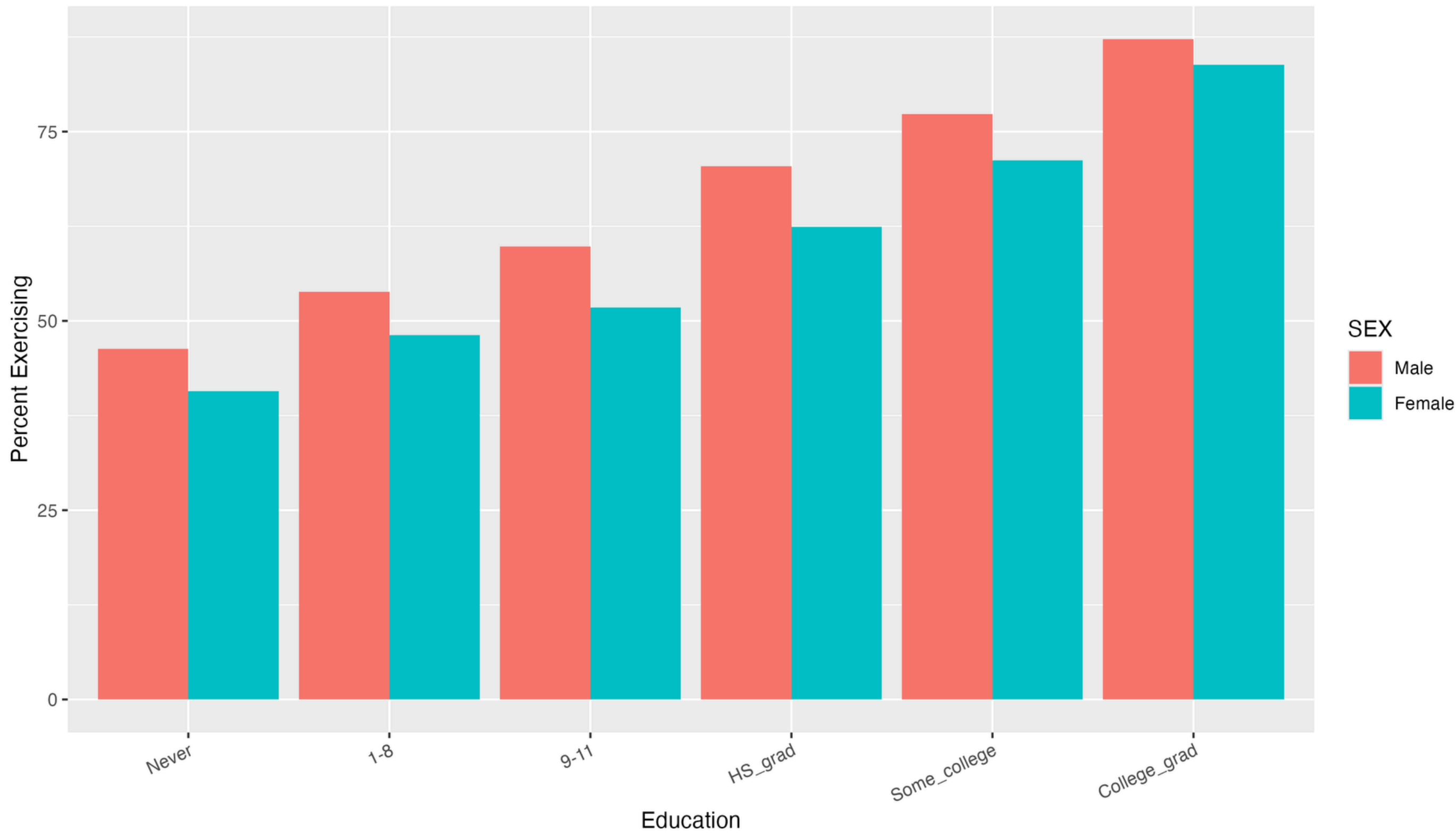
Logistic Model Coefficients (log-odds)



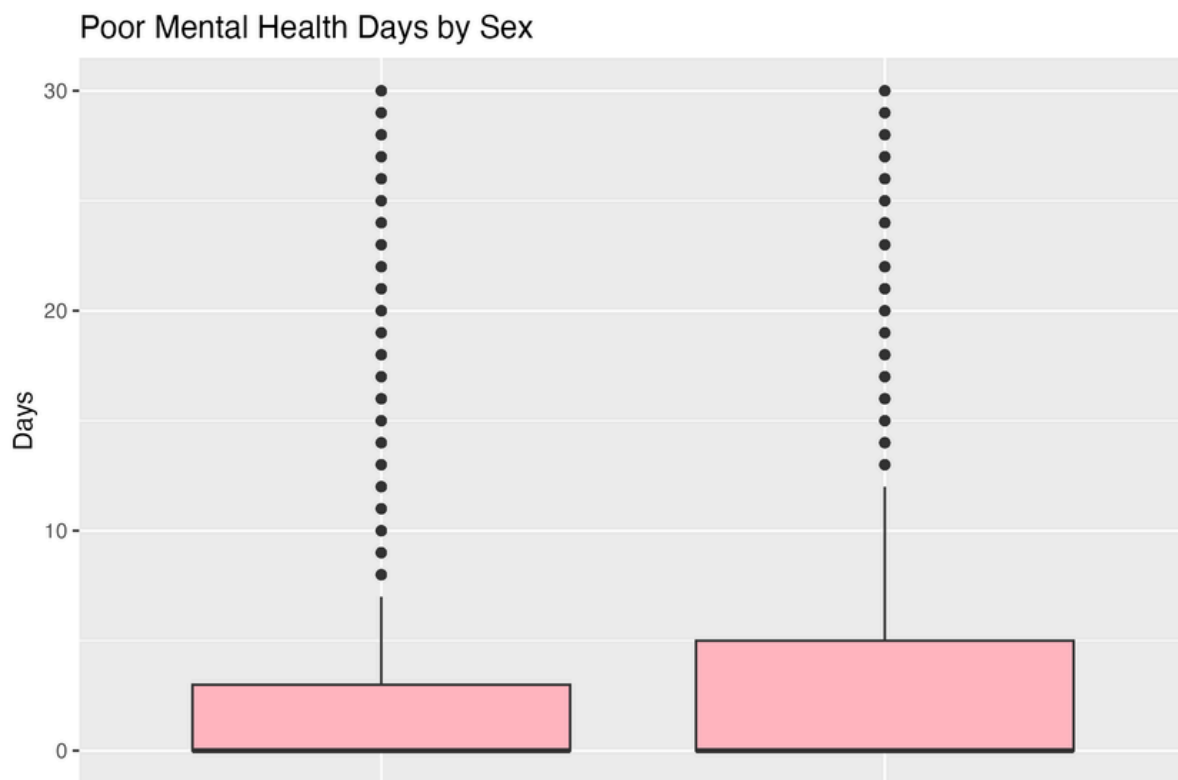
Visualization #3

Grouped bar chart (exercise × education × sex)

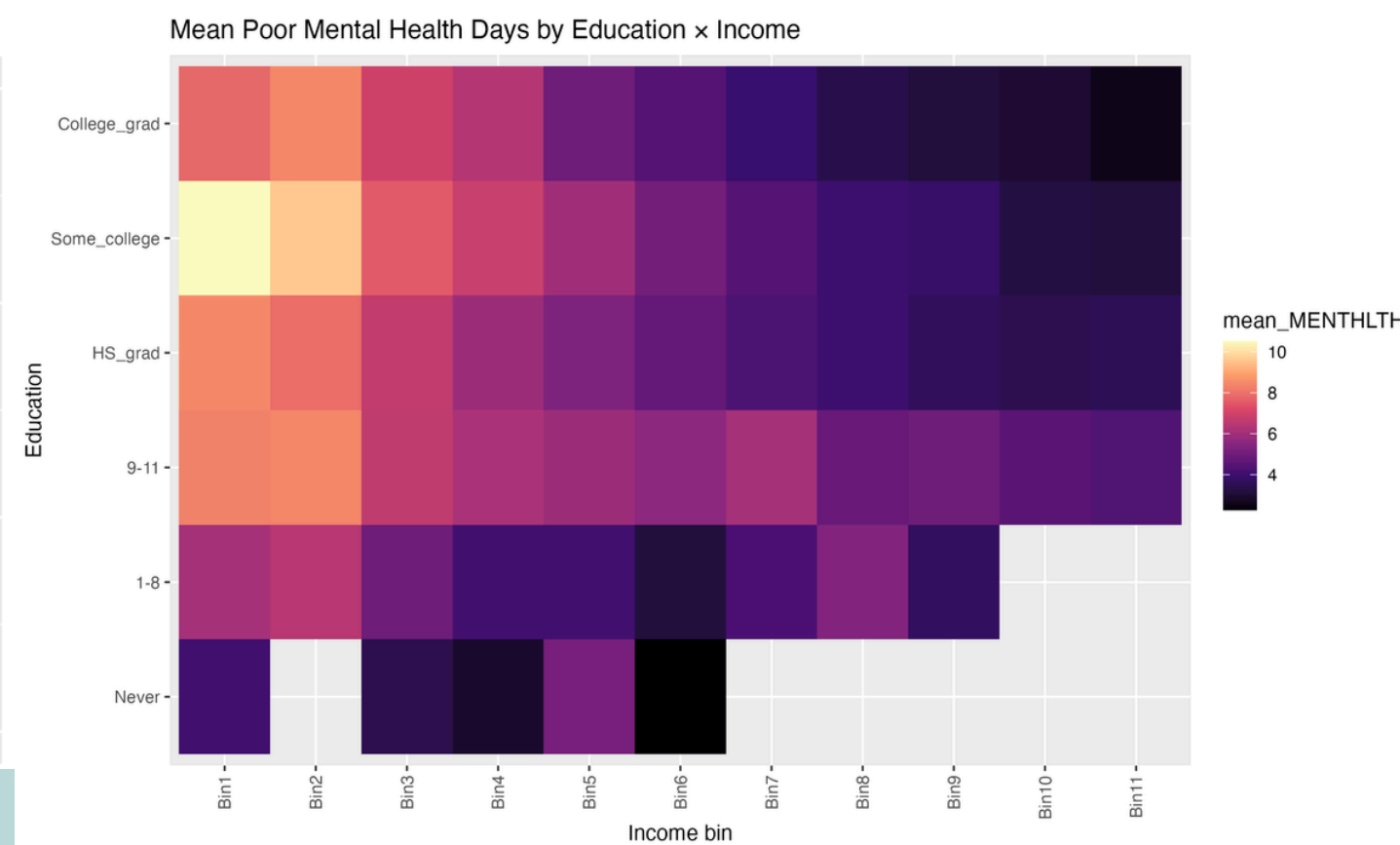
Percent Exercising in Past 30 Days by Education and Sex



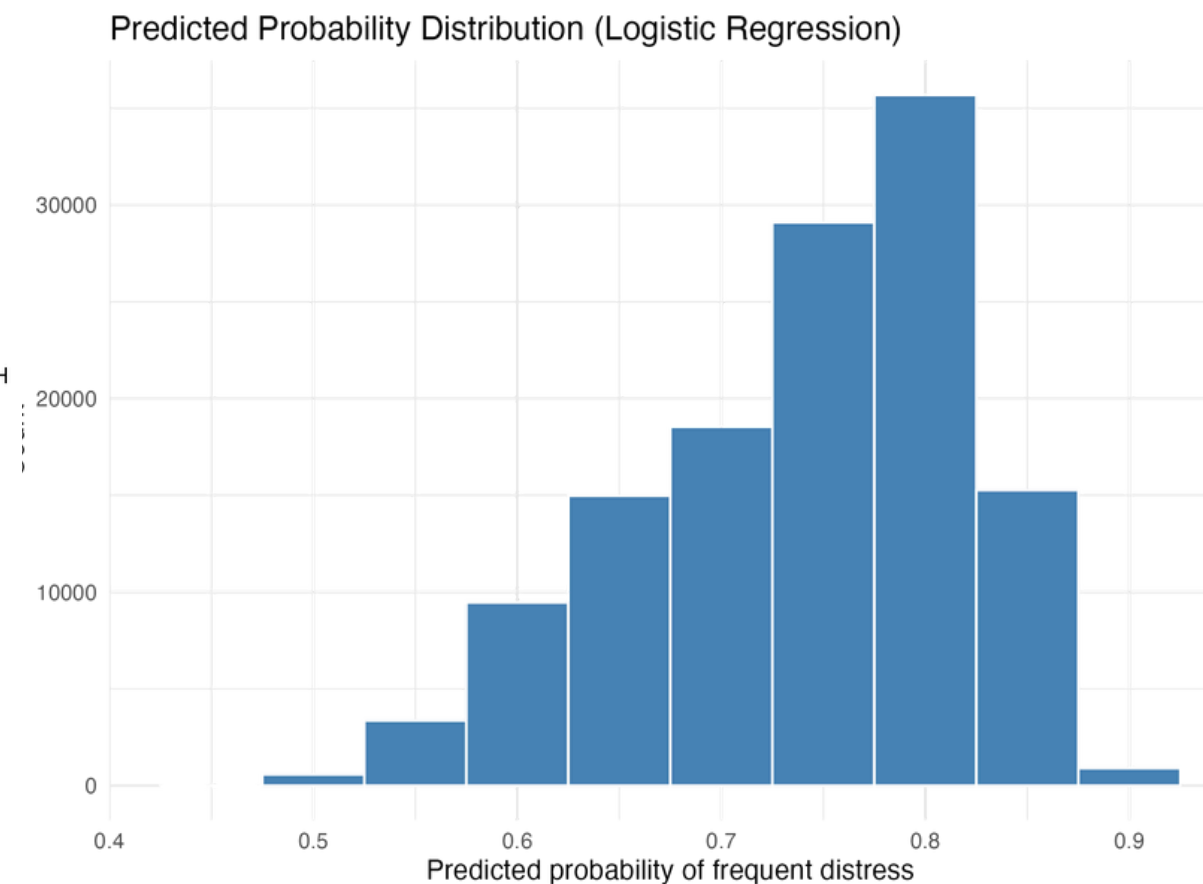
Additional Analysis



Stratified analysis (sex & age groups)



Interaction effect interpretation



Any additional models

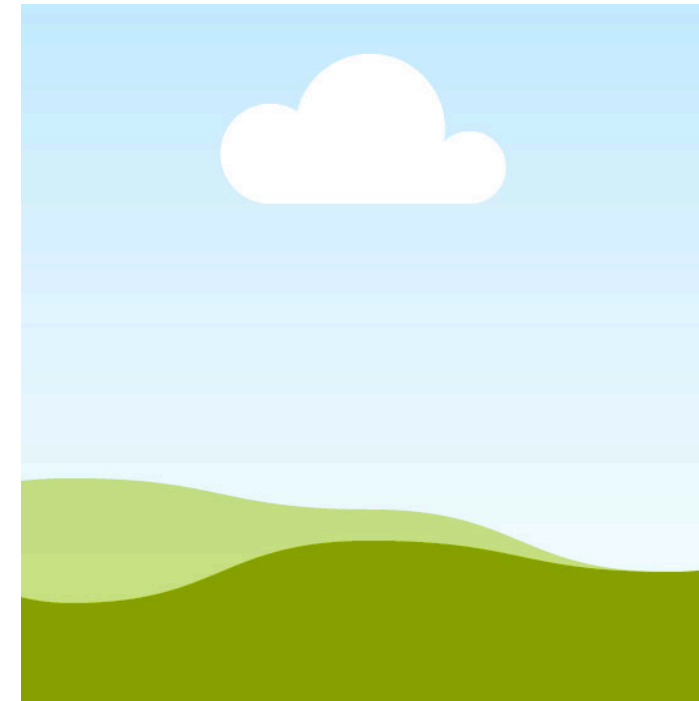
Findings / Results

Behavioral variables (exercise, smoking, alcohol) remained significant even after controlling for demographics.



Conclusion

- Behavioral & socioeconomic predictors matter
- Regression provides interpretable insights
- BRFSS is strong but limited by self-report



Future Work

- SHAP, LIME
- Survey-weighted ML
- Longitudinal models



References

APA References link (Google Docs):

[Click here](#)

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Thank You!
